

UPINGTON, South Africa

WMO: 684240

Lat: 28.4111S

Lon: 21.2642E

Elev: 2756

StdP: 13.29

Time Zone: 2.00 (E02)

Period: 94-19

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	32.2	35.2	4.9	7.8	61.6	8.3	9.3	63.7	25.8	66.8	22.9	68.5	3.2	260	0.450

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
1	27.8	103.2	63.8	100.7	63.3	98.8	62.9	71.1	85.9	70.0	84.9	68.9	84.8	10.2	280

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	
67.9	113.4	75.2	66.0	106.4	75.3	64.2	99.7	75.9	36.8	85.5	35.8	83.6	34.8	84.7	78.6

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
1%	2.5%	5%	Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
			Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max		
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
22.4	19.3	16.8	DB	27.5	107.1	2.2	1.8	25.9	108.4	24.6	109.4	23.4	110.4	21.8	111.7	
			WB	23.9	73.1	1.6	2.5	22.8	74.9	21.8	76.4	20.9	77.8	19.8	79.6	

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec		
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(6)	Temperatures, Degree-Days and Degree-Hours	DBAvg	69.8	82.5	82.2	78.1	70.1	62.7	55.5	55.2	58.7	65.3	71.9	76.0	80.5	(6)
(7)		DBStd	11.70	4.81	4.63	5.53	6.71	6.88	5.95	6.62	7.72	7.66	7.34	6.75	5.11	(7)
(8)		HDD50	62	0	0	0	0	4	17	28	12	1	0	0	0	(8)
(9)		HDD65	1099	0	0	1	29	125	289	309	226	90	22	8	0	(9)
(10)		CDD50	7296	1009	901	870	603	399	181	188	281	460	678	781	945	(10)
(11)		CDD65	2858	544	481	406	182	54	3	4	31	99	235	339	480	(11)
(12)		CDH74	39247	7502	6327	4910	2114	831	138	166	691	1743	3377	4754	6694	(12)
(13)		CDH80	21429	4511	3705	2660	908	216	6	14	195	792	1760	2686	3976	(13)
(14)	Wind	WSAvg	7.6	8.1	7.7	7.2	7.0	6.8	6.8	7.3	7.5	7.5	8.1	8.4	8.5	(14)
(15)	Precipitation	PrecAvg	7.8	1.2	1.5	1.5	0.9	0.4	0.2	0.1	0.1	0.2	0.4	0.6	0.7	(15)
(16)		PrecMax	19.7	6.3	6.0	5.3	4.4	1.9	1.1	1.1	1.1	0.9	2.9	3.2	3.0	(16)
(17)		PrecMin	2.1	0.0	0.0	0.1	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	(17)
(18)		PrecStd	3.4	1.4	1.5	1.3	0.9	0.5	0.3	0.2	0.2	0.2	0.5	0.6	0.9	(18)
(19)	Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	106.0	104.4	102.2	96.4	88.1	80.6	82.0	88.5	97.2	100.9	103.6	104.8	(19)
MCWB			64.3	65.7	64.8	62.4	57.9	54.1	53.2	56.2	59.4	61.3	62.1	63.3	(20)	
2%		DB	102.9	101.8	98.8	91.8	85.3	77.4	78.4	84.8	93.1	97.0	100.1	102.0	(21)	
		MCWB	64.2	64.7	63.7	61.6	57.2	53.1	52.5	54.9	58.1	60.4	61.5	62.9	(22)	
5%		DB	100.4	98.9	96.4	89.3	82.4	75.2	75.3	82.0	88.7	93.7	96.9	99.0	(23)	
		MCWB	63.5	64.8	63.6	61.2	56.5	52.1	51.6	53.7	56.5	59.4	60.4	62.1	(24)	
10%		DB	97.4	96.6	93.1	85.8	79.0	71.8	71.8	77.6	84.4	90.0	93.5	96.6	(25)	
		MCWB	63.8	64.6	63.5	60.7	55.8	51.0	50.5	52.4	54.9	58.4	59.9	61.8	(26)	
(27)	Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	72.9	72.8	71.5	68.5	62.8	57.3	56.2	59.3	63.3	65.9	68.8	70.0	(27)
MCD B			86.7	87.8	83.3	82.6	75.8	75.8	74.2	81.3	85.1	85.6	85.2	86.4	(28)	
2%		WB	71.1	71.1	69.7	66.1	60.2	55.0	54.2	57.0	60.8	64.0	66.0	68.2	(29)	
		MCD B	85.7	86.7	84.1	79.3	76.4	73.8	73.9	79.5	83.5	84.7	86.8	86.1	(30)	
5%		WB	69.9	70.0	68.4	64.6	58.5	53.2	52.6	55.0	58.6	62.4	64.2	66.8	(31)	
		MCD B	84.5	85.3	83.4	80.1	76.3	72.1	73.0	78.5	83.2	85.8	86.3	86.0	(32)	
10%		WB	68.6	68.9	67.0	63.0	56.9	51.6	51.0	53.0	56.4	60.4	62.0	65.3	(33)	
		MCD B	85.1	85.0	82.7	79.8	75.3	69.7	70.5	75.3	80.8	85.6	86.7	85.9	(34)	
(35)	Mean Daily Temperature Range	5% DB	MDBR	27.8	26.9	27.1	27.2	28.9	29.9	30.0	32.1	32.4	30.7	30.1	29.5	(35)
MCDBR			32.0	30.6	31.1	30.0	31.7	33.5	33.6	36.1	37.9	35.1	35.0	32.9	(36)	
MCWBR			11.7	11.0	11.0	11.0	13.1	16.2	16.2	17.2	17.4	14.6	13.7	12.6	(37)	
5% WB		MCD BR	23.2	22.9	23.7	24.1	27.0	30.0	30.0	32.3	32.3	28.9	27.5	26.4	(38)	
		MCWBR	8.4	7.8	9.0	9.8	11.8	15.1	15.3	15.8	15.4	12.7	12.1	10.4	(39)	
(40)	Clear-Sky Solar Irradiance	taub	0.334	0.345	0.337	0.312	0.281	0.265	0.270	0.304	0.338	0.354	0.315	0.318	(40)	
(41)		taud	2.475	2.450	2.455	2.502	2.570	2.584	2.550	2.413	2.302	2.301	2.475	2.495	(41)	
(42)		Ebn at Noon	320	311	303	296	292	291	294	294	298	304	324	325	(42)	
(43)		Edh at Noon	37	37	35	30	25	23	25	32	40	43	37	37	(43)	
(44)	All-Sky Solar Radiation	RadAvg	2613	2351	2054	1639	1339	1178	1284	1569	1950	2325	2633	2735	(44)	
(45)		RadStd	118	139	84	59	42	51	39	56	67	62	83	95	(45)	

Historical Trends

	DBAvg	Heating		Cooling			Degree-Days			
		99% DB	99% DP	1% DB	1% WB	1% DP	HDD50	HDD65	CDD50	CDD65
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)
(46) Station Only	N/A	N/A	N/A	N/A	-0.81	N/A	N/A	N/A	N/A	N/A
(47) Regional (1 neighbor)	N/A	N/A	N/A	N/A	-0.81	N/A	N/A	N/A	N/A	N/A

Nomenclature: See separate page